Towards Semantically-Aware End-of-Utterance Detection for Conversational AI Systems

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Abstract

A conversational voice agent has to decide, in real time, when a caller has finished speaking. On a French medical telephone line that decision is delicate: ending a turn too early cuts a patient off part-way through a spelled name or a dictated number, while ending it too late makes the exchange feel slow. Systems that rely on silence alone hear sound but not meaning, and cut off utterances that are grammatically incomplete.

This dissertation asks whether adding a lexical, meaning-aware signal improves that decision, and how that probability should become an endpoint. Two off-the-shelf models, an acoustic end-of-turn detector and a French language model reading the partial transcript, are fused into one probability; that probability is calibrated; and the operating point is chosen, in the offline evaluation, from an explicit interruption-versus-latency cost rather than from a timer. No base model is fine-tuned in the pipeline that produces this result, and no large language model sits in the decision loop.

On a public French benchmark the fusion raises the per-span AUC from 0.814 to 0.897, a gain of 0.083 (95 % confidence interval $[0.063, 0.104]$, $p < 1/3000$, bootstrap resampled by call). Across the leaderboard, calibration proves a field-wide blind spot: every external detector evaluated is miscalibrated and none reports it. Validation on 636 production medical calls is negative, the acoustic branch saturating on 8 kHz telephony and weak supervision capping the lexical signal, marking where the approach does not transfer. The contribution is a calibration-first, cost-aware method for composing existing models, with its limits measured.

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Chapter 1

Introduction

The rise of large language models (LLMs) has brought with it the voice agent: a system that can hold a conversation and answer a request, including in demanding fields such as healthcare. This is the setting for the present project, which was carried out at Tennor, a company that builds voice agents for healthcare, able to identify patients, answer their requests, and interact with the software used across the medical sector.

These agents share a design flaw. Because they are built to respond as quickly as possible, they tend to cut the caller off when a sentence is delivered in several pieces, for example a spelled-out detail, a telephone number given in blocks, or a pause for thought.

A person handles these pauses easily, from the meaning of the words and the speaker's intonation. A machine that listens only for silence does not. The central problem of this dissertation is therefore the following: how can the system tell that a caller has not finished speaking, so that it does not interrupt them, lose information, and degrade their experience? The decision has to be made in real time after every silence, and it rests on an uncomfortable trade-off: ending the turn too early talks over the caller, while waiting too long makes the exchange feel slow.

The title of this dissertation says end-of-utterance; the literature and the benchmark used here say end-of-turn. They name the same event, the moment a speaker has finished and the floor is free, and the two are used interchangeably below.

The existing system already separates speech from silence with a voice activity detector (VAD) and ends the turn after a fixed timeout. It listens to the sound, but not to the meaning. My idea is to add a semantic signal: a language model reads the partial transcript and estimates whether the words form a complete utterance. This signal is fused with the acoustic one into a single probability,

that probability is calibrated, and a stop-or-wait decision is then derived, in the offline evaluation, from an explicit cost: the price of an interruption set against that of a delay, rather than a simple timer.

Fusing an acoustic signal with a lexical one is not, in itself, new. The contribution of this project is narrower: a composition of off-the-shelf models, with no joint training and no large language model in the decision loop; a probability that is validated before it is trusted; and a decision derived from an explicit cost. The whole study is grounded in a real case of French medical telephony, with an in-domain dataset gathered by instrumenting the production system.

How that claim came to be scoped so narrowly is part of the work. When I chose this project and had it approved, in April 2026, systems fusing an acoustic and a lexical signal existed, as Chapter 2 sets out, but none I could find treated its output as a probability to be calibrated or derived its endpoint from a stated cost.

The critical review I submitted on 12 June, of a predictive-endpointing paper (Zink et al. 2024), ended by proposing the three things that became this dissertation: the fusion, the calibrated cost-sensitive decision, and the design that never waits on the transcript. Five days later LiveKit released a turn detector and an open benchmark that overlapped with the original angle. Rather than reproduce it, I narrowed the claim to the two things the released systems still do not do. Their benchmark is also the one this dissertation evaluates on.

On a public French benchmark the fusion raises the per-span AUC from 0.814 to 0.897, a statistically significant gain obtained without fine-tuning either base model. In domain, however, the same off-the-shelf pipeline does not transfer, and understanding why turns out to be the more instructive result of the two.

1.1 Aims and objectives

I set the project four objectives.

- **O1.** Build a modular, real-time endpointing pipeline (VAD, acoustic branch, lexical branch, fusion, decision) whose decision never waits on the transcript.
- **O2.** Fuse an off-the-shelf acoustic detector and a French language model, and test whether the fusion beats the acoustic model alone on a public benchmark.
- **O3.** Calibrate the fused score and choose the operating point from an explicit cost (interruption against latency) rather than a fixed timeout.

- **O4.** Instrument the production agent to build a weakly labelled in-domain evaluation set, and validate the method in domain.

Chapter 2

Background and Related Work

2.1 The turn problem inside a conversation between two humans

Turn-taking is the way two speakers alternate in holding the floor. For a machine, the difficulty is to decide, at every short pause in the exchange, whether to start speaking because the turn has ended, or to keep waiting because the other party has not finished. That decision rests on a trade-off that Section 3.1 states formally, and no setting escapes both of its errors.

In most dialogue systems the decision is taken from silence alone: the system waits for a pause of fixed length and then treats the turn as over. The organisation of turn-taking has been studied since Sacks, Schegloff and Jefferson (1974), and the rule is simple and fast, but it conflates two very different situations, a genuine end of turn and a pause in the middle of an utterance, because both present themselves as silence. In medical telephony this confusion is both frequent and costly: a patient spelling a name or dictating a number produces silences that are not ends of turn.

These are precisely the cases on which models that listen only to sound fail.

Having no access to the meaning of the words, they cannot separate a finished sentence from a spelling sequence or a number still in progress. A perceptual study on French shows that the two channels are not interchangeable (Uro et al. 2024). Inter-pause units drawn from the REPERE corpus were annotated for terminality and presented to participants either as audio or as text, with the end of the unit truncated by up to three prosodic words; participants then judged whether the floor could be taken. The audio condition allowed a turn end to be anticipated at least three words before it arrived, the text condition considerably less.

That result ranks the two channels, and ranks them in favour of audio, which is the baseline this project sets out to improve. What it establishes is separability: the same boundary judged from sound and from words produces different judgements. Whether the difference combines into something better than either channel alone is an empirical question, and Section 5.2 answers it.

2.2 Tools and technologies

The system reuses several existing components. Each is introduced with the reason it was chosen, because for a project whose contribution is the composition of off-the-shelf parts the selection is itself a design decision.

Voice activity detection is handled by Silero VAD (Silero Team 2024), an open-source package with no accompanying academic publication, which labels each short audio frame as speech or silence. It is the entry point of the pipeline: it triggers an evaluation at every transition from speech to silence. It was chosen because Tennor had already selected it for the production agent on the basis of an internal comparison; I did not repeat that comparison, and I take the choice as given. Using the same detector keeps the evaluated pipeline aligned with the deployed one.

The acoustic end-of-turn estimate comes from Smart Turn v3, a model released by Pipecat and Daily (Pipecat AI 2026), likewise without an academic publication and documented only by its model card and a technical blog post. It uses the encoder layers of Whisper Tiny (Radford et al. 2023) with a single linear classification head on top, for roughly eight million parameters. From the recent audio it produces a probability that the turn has ended, without reading any text. This project uses version 3.2 in its CPU export. Versions 3.1 and 3.2 improved the handling of short utterances and robustness to noise, and the exact version is stated because the reported figures depend on it.

Smart Turn was chosen for three reasons, and the absence of alternatives is one of them. It is open source under a permissive licence, so it can be run, inspected and redistributed inside a dissertation; it is the model the open turn-taking ecosystem has converged on, so comparing against it compares against what practitioners actually deploy; and it runs on CPU at eight million parameters, which the real-time constraint of Section 3.1 requires. No other openly available end-of-turn detector met all three at the time of selection.

The lexical estimate comes from CamemBERT (Martin et al. 2020), a French BERT-style language model applied to the partial transcript. Unlike the acoustic signal, it judges completeness from the meaning of the words. It was chosen

because the target language is French and CamemBERT is pre-trained on French rather than adapted to it from a multilingual model, which matters for a judgement that turns on grammatical completeness; and because it is small enough to fine-tune on a modest dataset, which Section 3.4 takes up.

Continuous transcription is provided by Azure Speech, which emits partial hypotheses as speech unfolds. The reason here is continuity rather than superiority: it is the recogniser the production agent already uses, so the transcripts the lexical branch reads in evaluation are the ones it would read in deployment. Substituting a different service would have measured a pipeline that does not exist.

Evaluation relies on eot-bench (LiveKit 2026), which its authors describe as the first open collection of genuine human-to-agent turns with aligned audio and textual context, covering fourteen languages of which French is one. Rather than fixing a single threshold, the harness sweeps the policy jointly over threshold, action delay and timeout and reports the frontier between latency and cut-ins. It is an industrial artefact rather than a peer-reviewed publication, released under Apache-2.0 in June 2026 alongside the LiveKit Turn Detector v1, three weeks before implementation began here.

One caveat qualifies Chapter 5. Its authors omit scalar ranking metrics such as the area under the receiver operating characteristic curve (AUC) from their comparison reports, treating them as diagnostic only. This dissertation reports per-span AUC as its headline figure because the question here is whether a lexical signal improves the underlying *score*, which a policy sweep conflates with the timing applied on top of it. Both views are reported throughout.

2.3 Calibration and cost-sensitive decision

Neither of the two bodies of method underpinning the decision layer is specific to turn-taking.

The first is probability calibration. A classifier is calibrated when its scores match observed frequencies at every level: among the cases it assigns a probability of p, a proportion close to p should be positive. Taking $p = 0.8$ as an illustration, roughly eight in ten of the cases it calls 0.8 should turn out positive, and the same must hold at 0.2, at 0.5 and everywhere else. Modern neural networks are systematically miscalibrated and typically overconfident; Guo et al. (2017) attribute this to model capacity and to training choices such as batch normalisation, and show that accuracy and calibration can move in opposite directions. Calibration is therefore not implied by good discrimination. A model can rank correctly, and so obtain a high AUC, while the numbers it emits do not correspond to any real

rate of occurrence: a score of 0.9 might be right nine times in ten, or six, or four. The ordering is informative, the value is not.

The standard remedies are post-hoc and monotonic. Platt scaling fits a sigmoid; isotonic regression (Zadrozny and Elkan 2002) fits an arbitrary non-decreasing step function, more flexible but hungrier for held-out data (Niculescu-Mizil and Caruana 2005). Being monotonic, neither changes the ranking, and so neither changes the AUC: they correct the values, not the ordering. Calibration quality is measured by the expected calibration error, or ECE (Naeini, Cooper and Hauskrecht 2015; Guo et al. 2017), defined in Section 3.7.

The second is cost-sensitive decision making. Once a probability is trustworthy, where to act is not a free parameter but a consequence of the relative cost of the two errors: Elkan (2001) shows that under an asymmetric loss the operating point follows from the cost ratio rather than being set by hand. The rule used here decides on the evidence available at one instant. The richer alternative lets observations accumulate and weighs acting now against paying to observe more; the canonical formulation is Wald's sequential probability ratio test (Wald 1945), which Chapter 6 returns to as future work.

A third strand is relevant to the in-domain dataset rather than the decision. Weak supervision replaces manual annotation with cheap, noisy labelling heuristics, an approach systematised by Ratner et al. (2017). It buys scale and, in a clinical setting, avoids a person having to listen to call content, at the cost of label noise that bounds the signal any model can recover from the data. That bound explains the in-domain result of Section 5.7.

2.4 Related work on end-of-turn detection

Work on end-of-turn detection can be grouped by the signals it uses.

Purely acoustic approaches decide from sound, prosody and silence, without reading the words. They are fast and independent of transcription, but blind to meaning, and so cut off utterances that are grammatically incomplete. Smart Turn, the baseline of this project, belongs to this family. A more elaborate member is Zink et al. (2024), which trains a Conformer to transcribe before an utterance finishes and reads the decoder's cross-attention at the end-of-sentence token to locate the endpoint, so no separate language model is needed. It is evaluated offline on pre-segmented utterances, however, and never inside a running dialogue system, so it is not shown that the timing it saves survives streaming constraints. That gap shaped the present design.

Purely lexical approaches read the transcribed text and judge whether the utterance is complete. Phoenix-VAD (Wu et al. 2025) is the most elaborate of them and takes the opposite route to the one adopted here, putting a large language model inside the endpoint decision itself. They supply the meaning the acoustic signal lacks, but they inherit a dependency that shapes both the design of Chapter 3 and the failure reported in Section 5.7. A lexical branch is only as good as the transcript it reads, and a streaming transcript is late, unstable and imperfect: partials arrive behind the audio, are revised as context arrives, and degrade on narrowband telephone speech. Waiting for a stable transcript buys accuracy with the latency endpointing is trying to save; reading whatever is available reads a text that may be wrong. Neither horn is free.

Joint approaches process audio and text together. FastTurn (Wang et al. 2026) unifies streaming connectionist temporal classification (CTC) decoding with acoustic features so that a decision can be taken early from partial observations while semantic cues are preserved. It performs well but is heavy and presupposes joint training of its components. JAL-Turn (Yang et al. 2026) combines pre-trained acoustic representations with linguistic features through a lightweight joint model built on a frozen automatic speech recognition (ASR) encoder, so that it runs in parallel with recognition and adds no latency. Unlike FastTurn it does not require the two modalities to be retrained together, and its labels are extracted automatically from large-scale dialogue corpora.

X2-Turn (Fu et al. 2026) goes further, placing a frame-synchronous turn state head alongside the ASR head on shared streaming features.

On the benchmarking side, the LiveKit detector and test set provide the policy-sweep evaluation framework this project adopts. Other work remains purely acoustic: endpoint anticipation (Udupa et al. 2026) bypasses ASR to predict a turn ending in advance, up to roughly 2.5 seconds ahead, so that the rest of the pipeline can be started sooner. Its object is not the choice of a decision threshold but early prediction, and the cost it seeks to reduce is the speculative computation wasted when the pipeline is launched too early and then abandoned.

The closest neighbour is Easy Turn (Li et al. 2025), an open-source modular detector that also integrates acoustic and linguistic information and also positions itself against Smart Turn.

Beyond that the two diverge on every axis that matters. Easy Turn trains: its model is fitted on a purpose-built 1,145-hour corpus and predicts one of four dialogue states, from which a decision is taken by argmax. The present work trains no base model, fitting only a linear head on frozen embeddings and a four-parameter fusion above it, and emits a scalar probability rather than a class.

Easy Turn reports accuracy on its own test set; this work reports on a third-party

benchmark in a language subset it did not choose. And neither Easy Turn nor any system reviewed here reports whether its probability is calibrated, or connects its operating point to a stated cost. Fusion is done in both; what differs is that the output is treated as a probability to be validated before a threshold placed on it can mean anything.

Table 2.1 summarises the comparison.

Table 2.1: Representative recent end-of-turn detection systems and where this work sits. "Calib." records whether the system reports any measurement or correction of the calibration of its own output probability.

System	Signals	Training	Decision rule	Calib.	Evaluation
Easy Turn (2025)	acoustic + linguistic	trained, 1,145 h	argmax over 4 states	no	own test set
Phoenix-VAD (2025)	LLM over text	LLM fine-tuned	LLM in the loop	no	own test set
JAL-Turn (2026)	acoustic + linguistic	head only, frozen ASR	fixed threshold	no	multilingual
FastTurn (2026)	acoustic + CTC	joint training	early, on partials	no	own test set
X2-Turn (2026)	shared ASR features	joint, dual-head	per-frame state	no	own test set
Endpoint antic. (2026)	acoustic only	trained	per-horizon threshold	no	anticipation only
LiveKit v1 (2026)	text (+ VAD)	trained detector	swept policy	no	14-language benchmark
This work	acoustic + lexical	**frozen bases**; linear head + fusion	**cost-derived, on a calibrated probability**	**yes**	French benchmark + medical domain

The fusion mechanism itself, combining an acoustic signal with a lexical one, and the latency-versus-cut-in framing are not new, and this project claims to have invented neither. The field is moving fast enough that every one of the seven systems in Table 2.1 was published in the eleven months to August 2026, and the most recent of them appeared three weeks before this chapter was written.

2.5 The gap addressed

Three gaps emerge from this review. Almost all of the work above appeared after this project was scoped (Chapter 1), so these are the gaps that remain *after* that shift, not the ones the project was originally scoped against.

First, none of the systems reviewed reports an explicit calibration of its probability, neither an expected calibration error nor any recalibration, and none relates its decision threshold to a cost of interruption against latency: eot-bench sweeps thresholds and some vendors expose confidence scores, but the probability itself is neither measured nor corrected, so a chosen threshold has no precise meaning.

Second, Table 2.1 shows the field moving steadily towards larger jointly trained systems. The opposite end of the design space, where frozen components are merely composed, is little studied. Third, the French medical telephony domain, where semantically incomplete utterances (spelling, digits, dates) are both frequent and consequential, is not addressed. These are the three gaps this dissertation sets out to close.

Chapter 3

Design and Trade-Off Analysis

3.1 The central trade-off: latency against interruption

Every decision in this chapter serves the trade-off introduced in Chapter 2, which I state formally here. At each micro-pause, ending the turn too early cuts the patient off, an interruption of cost C_{cut}, while waiting too long delays the reply, latency of cost C_{lat}. Improving one degrades the other, and no setting escapes both. Designing the system therefore comes down to deciding, at every instant, which side of that trade-off to sit on, and deciding it for a stated reason. Each section below sets out one decision and the trade-off it resolves.

3.2 Which data trains, and which data validates

The project started from the assumption that Tennor's production transcripts would be the training data and that the evaluation would be built around them. I reversed that in late July, and because the reversal shaped everything downstream it belongs here rather than in the implementation chapter.

Licensing forced it, and so did the labels. The eot-bench harness is Apache-2.0 and its dataset CC-BY-4.0, both usable for research and derivative work; the production data is personal data that cannot leave Tennor's infrastructure at all (Appendix A), so anything built on it alone would be a result nobody else could check. The two also differ in what they can support: the benchmark carries human-verified turn boundaries, while the production set is labelled by weak supervision, from whether the caller resumed speaking, which is cheap and scalable but noisy by construction.

So I inverted the roles. The public benchmark carries the headline experiments (fusion, calibration, cost), and the production data becomes an in-domain validation set, asked only the narrower question of whether the method transfers.

The cost of that inversion is real and worth naming. The headline result is obtained on generic French rather than on medical French, so the domain that motivates the project is not the domain the main claim is measured on. And a weakly labelled validation set can only return a weak answer, which is precisely what Chapter 5 reports. What it buys is that every headline number in this dissertation can be reproduced by anyone with a Hugging Face token, and none of them rests on data that nobody outside Tennor can see.

3.3 Late fusion rather than a jointly trained model

The first modelling decision was how to combine the acoustic and the lexical signal. One option is a single model trained jointly on audio and text. It can be more accurate, because it learns the interactions between the two modalities, but it is heavier to run, it demands aligned joint training data, and it locks both signals inside one box that cannot be replaced piecemeal. The alternative, and the one I took, is late fusion: each signal comes from an independent off-the-shelf model, and only their combination is learned.

I may lose some accuracy that way. What I gain is modularity, since each component can be swapped on its own, along with speed and simplicity, which matches the constraint the project set itself of composing existing models without retraining them.

3.4 The lexical model: frozen embeddings rather than fine-tuning

The lexical signal can be obtained in two ways. CamemBERT can be fine-tuned on Tennor medical French, which produces a *medical-fitted* model closely adapted to that domain but out of distribution on generic French, where it degrades the decision rather than improving it. Section 5.2 measures this. Or CamemBERT can be left frozen and used purely as a feature extractor, with a small classifier on top, which generalises better and cannot overfit the way a fine-tuned encoder can.

The two are named for the data they were fitted on in the chapters that follow: *benchmark-fitted* is the frozen encoder under a head fitted on the benchmark,

and *medical-fitted* is the encoder fine-tuned on medical French. I kept both, for different paths. The frozen encoder with a head fitted on the benchmark, *benchmark-fitted* in the tables of Chapter 5, serves the benchmark evaluation. The medical-fitted model serves the in-domain path and the live demonstration, where medical French is what it will actually see.

3.5 A decision that never blocks on the transcript

One constraint governs the whole pipeline: the decision has to stay fast, while streaming transcription can and does fall behind. I could wait for the most complete transcript available, paying added latency and risking a stall, or read whatever partial transcript exists at the instant of the decision, as a snapshot, accepting that it may lag the audio.

I took the second. The decision never waits on the recogniser: the transcript is read as whatever snapshot exists at that instant. The two model calls are issued concurrently and both are awaited, so the added latency is the slower of the two rather than their sum. Added latency is bounded by construction, and the price is a text that is sometimes slightly incomplete. That is acceptable, because the acoustic signal is always available and the fusion degrades gracefully when the lexical branch has little to say. This is the design answer to the ASR dependency set out in Section 2.4.

3.6 The form of the fusion

The fusion takes two forms, and the difference between them runs through the rest of this dissertation. A rule-based form applies a bilateral veto and then a fixed average; it needs no data to fit, so it was the committed deliverable and it is what the system runs (Section 4.4). The two operations are sequential rather than combined: the veto is a gate, and when either signal falls below its threshold the decision is to wait and the average is never computed. Only when both signals clear the gate are they averaged, and the result is then mapped onto how much trailing silence to demand. A learned form fits the combination, and it is the one every result in Chapter 5 is measured on. The rule form is described with the implementation because it is an engineering artefact; the learned form is set out here because its shape is a design decision.

In the learned form the two probabilities are combined by logistic regression on p_{ac}, p_{lex} and their product. With weights $w_0 \dots w_3$,

$$z = w_0 + w_1 p_{\text{ac}} + w_2 p_{\text{lex}} + w_3\, p_{\text{ac}} p_{\text{lex}}, \qquad p_{\text{raw}} = \sigma(z) = \frac{1}{1 + e^{-z}}. \tag{3.1}$$

Here $p_{\text{ac}} \in [0, 1]$ is the probability of end of turn produced by the acoustic model and $p_{\text{lex}} \in [0, 1]$ the one produced by the lexical model, both read at the same micro-pause; z is the linear score they are combined into; σ is the logistic function; and $p_{\text{raw}} \in [0, 1]$ is the fused probability before any calibration. There are four weights because the model has four terms and no more: an intercept w_0, one weight per input signal, and w_3 on their product.

This form contains the rule-based one as a special case. The fixed average $0.5\,p_{\text{ac}} + 0.5\,p_{\text{lex}}$ is Equation 3.1 with $w_0 = 0$, $w_1 = w_2 = 0.5$ and $w_3 = 0$: the same shape, with the weights set by hand instead of fitted, and with the interaction term switched off. What fitting them adds is visible in the values Section 5.2 reports: the lexical branch receives roughly 2.2 times the weight of the acoustic one, so the equal split the rule form assumes is not what the data supports. The veto and the silence ladder run the other way: they are gating structure that Equation 3.1 has no counterpart for. The interaction term $p_{\text{ac}}p_{\text{lex}}$ lets the model treat the case where both signals agree as different from the sum of their separate contributions, which a purely additive form cannot represent.

I preferred logistic regression to richer combinations because it is light, its weights stay interpretable, so one can read off which of the two signals carries more of the decision, and its output is a probability that can then be calibrated. The fusion itself is small enough not to memorise a dataset of roughly a thousand spans, though the lexical head above it is not: a logistic layer on 768-dimensional frozen embeddings carries 769 parameters against some eight hundred training rows per fold, which is why every lexical and fusion figure in Chapter 5 is reported out-of-fold. The weights are fitted by cross-validation with folds split by call, so that no call appears in both training and test.

3.7 Calibrating before deciding

A structural decision is to treat the probability the system produces as a quantity that must be calibrated before it is used. The reason is direct. The final decision compares that probability against a threshold derived from a cost, and the comparison only means anything if the probability corresponds to a real frequency. As Section 2.3 sets out, a poorly calibrated model can rank cases correctly and still be systematically overconfident, which distorts any threshold placed on it.

Calibration is done by isotonic regression, fitting a monotonic g on a held-out split to minimise $\sum_i (g(p_i) - y_i)^2$. Because g is monotonic it leaves the ordering of predictions untouched, so the AUC does not move, and only the probability

values are corrected. Quality is measured by the expected calibration error,

$$\mathrm{ECE} = \sum_{b} \frac{n_b}{N} \left|\mathrm{acc}_b - \mathrm{conf}_b\right|, \tag{3.2}$$

The predictions are sorted into bins b by the probability they carry. For each bin, n_b is the number of predictions it holds, N the total number of predictions, conf_b the mean probability claimed within the bin, and acc_b the proportion of those cases that turned out positive. The ECE is therefore the average gap between claimed confidence and observed frequency, each bin weighted by how many predictions fall into it. The cost of the approach is that it consumes held-out data and can overfit on small sets, which I limit with a held-out split rather than reusing the fitting data.

3.8 Deciding from a cost rather than a fixed timeout

The last decision is the stopping rule. The existing system stops after a fixed silence, 500 ms say: simple, but arbitrary, and identical in every situation. Instead I choose the operating point from the trade-off of Section 3.1. Given the cost ratio

$$\lambda = \frac{C_{\mathrm{cut}}}{C_{\mathrm{lat}}}, \qquad \text{choose the policy minimising} \quad \lambda \cdot (\text{cut-in rate}) + (\text{latency}). \tag{3.3}$$

The units are easily lost. The cut-in rate is a fraction and the latency is in seconds, so λ is itself measured in seconds: $\lambda = 10$ says one interruption costs as much as ten seconds of extra waiting, not that it is ten times worse than a delay.

The operating point is then a consequence of a stated preference rather than a number fixed in advance. Equation 3.3 is the only form of the rule used here: the policy minimising the combined cost is selected directly, with no closed-form threshold derived from λ. A richer variant would fold the elapsed silence into the probability and decide sequentially, in the spirit of Wald (1945). I built it and it did not improve on the swept frontier (Section 5.6), so the calibrated single-instant rule stands, and the full sequential formulation is left to Chapter 6.

Chapter 4

System Architecture and Implementation

4.1 A modular architecture behind interfaces

The system is a Python package organised into modules, each behind an abstract interface, so that a component can be replaced without touching the rest (Figure 4.1). There is an audio source (`AudioSource`), a voice activity detector (`VAD`), a speech recogniser (`ASR`), the two end-of-turn branches (`AcousticEOU` and `LexicalEOU`), the fusion (`Fusion`) and the endpoint controller. This follows directly from the late-fusion choice of Section 3.3: the recogniser or the detector can be exchanged, a client to an existing microservice in place of the in-process model, without the decision logic changing at all.

The decision core is isolated from transport in a per-call session, `EouSession`. The session takes raw 16 kHz mono audio and the latest partial transcript, and emits a decision at each micro-pause. It is driven by an `Orchestrator`, which owns an audio source and a recogniser for the demonstrations and the evaluation harness, and by the WebSocket service, which drives a session directly. Both reach the same code, so there is one decision path and not two.

The session carries an explicit statefulness contract, which matters in production: the heavy models may be shared across concurrent calls, but the VAD holds state and must be instantiated per session.

4.2 The real-time path

The audio source emits frames of 512 samples at 16 kHz, that is 32 ms. In production the transport does not respect that framing, since telephony delivers 20 ms

frames, so the session accepts a chunk of arbitrary length and re-frames it internally to the fixed size the VAD expects. Each frame is written to a ring buffer holding the last eight seconds of signal, passed to the recogniser, and labelled speech or silence by the VAD. When a speech-to-silence transition lasts beyond 200 ms, an evaluation is triggered. That is the micro-pause at which the system decides to end the turn or to wait, and an internal latch guarantees it fires at most once per turn.

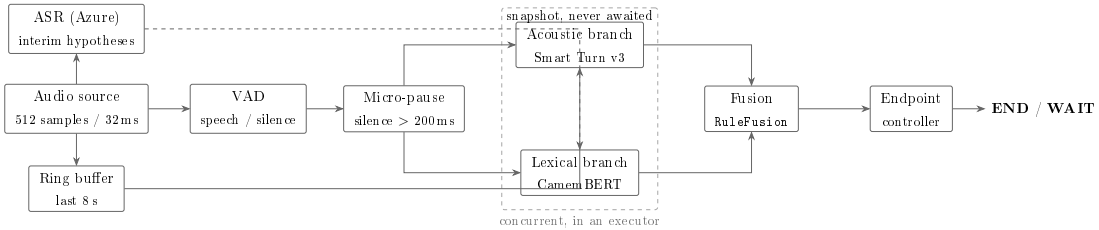


Figure 4.1: The decision pipeline. Every frame feeds the ring buffer, the recogniser and the VAD. A speech-to-silence transition beyond 200 ms triggers an evaluation, in which the two branches run concurrently off the event loop. The lexical branch reads the latest partial transcript as a snapshot, so the decision never waits on the recogniser; both model calls are awaited, and the added latency is the slower of the two. The fusion shown is `RuleFusion`, the variant the system runs; the learned form of Equation 3.1 substitutes behind the same interface but is evaluated offline only.

The session keeps two clocks. Silence duration is measured on an audio-sample clock, advanced by the frame duration, so a given recording always produces the same timing regardless of how fast the machine runs. Wall-clock time is used for one thing only: measuring how long the models took. A system timing silence by wall clock would behave differently on a loaded machine and on an idle one, and a replayed clip would not reproduce its own earlier result. This does not affect Chapter 5, whose figures come from stored predictions; it matters for the live pipeline and the clip-replay harness.

At the moment of evaluation the acoustic and lexical branches are computed concurrently, off the event loop in an executor, so that incoming frames continue to be processed. The partial transcript is read as a local snapshot and never awaited. That is the non-blocking property of Section 3.5 in concrete terms: the recogniser is off the critical path. The two model calls themselves are both awaited, so the added latency is $\max(\text{acoustic}, \text{lexical})$ and not their sum, and no timeout guards the lexical future.

4.3 The two branches and the recogniser

The acoustic branch applies Smart Turn v3 (version 3.2, CPU ONNX export) to the last eight seconds of the ring buffer and returns a probability that the turn has ended. The lexical branch applies CamemBERT to the latest partial and returns a completeness probability. The two variants of Section 3.4 sit behind the same interface, frozen embeddings with a classifier for the benchmark and the fine-tuned model for the domain and the live demonstration. A French heuristic guard (`FrenchSemanticEOU`) complements the model on the hard cases, detecting a spelling sequence still in progress, digits, and function words in final position.

Transcription reuses the production Azure service, with the same language and audio format. The one non-invasive addition is that it subscribes to the interim hypotheses as well as the final results, and those feed the lexical branch. Production itself is unchanged.

4.4 Fusion and the endpoint controller

`RuleFusion`, one of the two implementations behind the `Fusion` interface, applies a bilateral veto: if either signal falls below 0.25, that is, if either branch clearly reads mid-turn, the system waits. If both exceed 0.85 it endpoints immediately. Otherwise it averages the two and hands the result on. `LogisticFusion` substitutes for it without the calling code changing. It implements Equation 3.1 with one addition: a fifth term on the elapsed silence, normalised by the maximum-silence limit. The benchmark experiments of Chapter 5 use the four-parameter form of Equation 3.1 without that term, so that the reported gain is attributable to the two probability signals alone and not to timing information the acoustic baseline never receives.

One boundary should be drawn here. The learned fusion and the cost rule were staged from the outset, with the rule-based fusion as the committed deliverable, and neither reached the running path: `LogisticFusion` is instantiated nowhere outside a unit test, and no cost ratio appears anywhere in the runtime package.

The data strategy of Section 3.2 then fitted both the fusion and the calibrator on generic French, and Section 5.2 measures what a component fitted on the wrong domain costs. What the running system demonstrates is the pipeline, its real-time and non-blocking behaviour, and the capture that produced the in-domain set; the decision layer of Chapter 5 is established offline on stored predictions, and carrying it across is deliberately not claimed here.

The endpoint controller then converts confidence into a duration. Rather than a single threshold it maps confidence onto how much trailing silence is demanded:

200 ms above 0.85, 600 ms above 0.5, 1200 ms below. Confidence therefore buys speed, and a hesitant score is made to wait longer rather than being refused outright.

Above all of that sits a hard limit at 2000 ms, which deliberately overrides the veto. A caller who genuinely stopped mid-sentence, and whose text will never look complete, would otherwise be held indefinitely by the lexical branch. This is the one place where the design accepts a wrong cut over an unbounded wait. The reasoning is about failure modes: an interruption is recoverable in conversation, silence from the agent is not.

4.5 Demonstrations and evaluation tooling

Two demonstrations drive the same core. One runs at the command line, from the microphone or a wav file replayed as a stream; the other runs in the browser. Both print, at each decision, the acoustic, lexical and fused probabilities, the decision taken, the observed silence and the compute latency.

Evaluation rests on two tools: a local harness that replays labelled clips and computes accuracy, false positives, false negatives and latency, and an adapter that plugs the fusion into the eot-bench protocol so it can be compared with other systems on identical metrics.

The in-domain scoring path did not survive that route, and the reason cost me a day: the harness reads audio through the Hugging Face `Audio` feature, which now requires `torchcodec`, which does not load against `ffmpeg` 8. Rather than pin the toolchain backwards, a standalone scorer reads the caller wav files with `soundfile` and scores each pause with the same models and fold splitting. It produces the figures of Section 5.7.

4.6 Production instrumentation

To build the in-domain set, the production system was instrumented. Capture is gated behind an organisation-level feature flag. When active, the system records at each micro-pause the caller-side text only, the trajectory of the recogniser's interim hypotheses, timestamps, a reference to the audio, and resume or interruption flags. Labels come from weak supervision with no manual annotation: a pause followed by the caller resuming is a hold, a pause followed by the agent replying without a resume is an end of turn.

The resulting set is 709 calls placed on a single production organisation, of which roughly four in five are genuine inbound calls and the rest calls made by internal

testers to the same line. Of those, 636 contribute at least one scored micro-pause. Names, first names and ages are masked in the retained transcripts, and the data stays within Tennor's infrastructure. Because caller audio is retained and linked to a call identifier, it remains personal data and is treated as such throughout. Appendix A sets out the governance in full.

4.7 Testing and verification

The system is covered by 45 automated tests, exercising the VAD, the ring buffer, the fusion and endpoint logic, the lexical rules, the types and the metrics. Integration tests that need heavy models, credentials or audio hardware skip themselves when those are absent, so the suite runs anywhere.

Chapter 5

Results and Evaluation

5.1 Evaluation protocol

Evaluation runs through the eot-bench harness on the French subset: 400 turns and 1,054 silences, of which 400 are ends of turn and 654 are holds. Each of those silences is one micro-pause in the sense of Section 4.2: one moment at which the system must decide. The harness sweeps the endpointing policy and reports two measures. Per-span AUC says how well a model tells a silence that ends a turn from one that is only a pause. The latency and cut-in frontier says how long the system waits, and how often it speaks over a caller who has not finished.

Take one silence that ends a turn and one that does not. The per-span AUC is the probability that the model scores the first higher. 0.5 is a coin flip, 1.0 is a perfect ordering. All lexical and fusion figures are cross-validated with folds split by call, so no call appears in both training and test.

One methodological point conditions the comparisons. Three of the systems decide as the transcript arrives: the streaming recognisers Soniox, AssemblyAI and OpenAI GPT-Realtime 2. They cannot be scored on per-span AUC, because they score at a different moment from the one evaluated here. Their latency is reported and their AUC left out, since the number would not be comparable.

5.2 Fusion against the acoustic baseline

Table 5.1 gives AUC and latency, the latter at a 5 % and then a 10 % cut-in budget.

Table 5.1: Fusion results on the public French benchmark (eot-bench, 1,054 silences). *Benchmark-fitted* means the lexical head was fitted on the benchmark itself, cross-validated with folds split by call; *medical-fitted* means the lexical model was fine-tuned on Tennor medical French and applied here unchanged. The lexical branch was scored per span but never run through the policy sweep on its own, so it carries no latency figure. TF-IDF is term frequency-inverse document frequency, STT is speech-to-text.

Model	AUC	Latency @ 5 %	@ 10 %
Acoustic only (Smart Turn v3)	0.814	873 ms	608 ms
Lexical only (benchmark-fitted)	0.845	—	—
Fusion, medical-fitted lexical	0.789	850 ms	659 ms
Fusion, benchmark-fitted TF-IDF	0.872	759 ms	548 ms
Fusion, benchmark-fitted CamemBERT	**0.897**	**715 ms**	**510 ms**
ultraVAD	0.817	846 ms	613 ms
LiveKit v1-mini (commercial)	0.884	798 ms	541 ms
Deepgram Flux (commercial)	0.915	676 ms	353 ms
LiveKit v1 (commercial)	0.937	635 ms	376 ms
Soniox (streaming STT)	n/a	557 ms	463 ms

Fusion lifts the AUC from 0.814, for the acoustic model, to 0.897: a gain of 0.083. That is the comparison the objectives are stated against, but not the whole picture. The lexical branch alone reaches 0.845, so what fusion adds over the better of its two inputs is 0.052, not 0.083. The first number says what the deployed system would gain, the second whether combining the signals beats taking the stronger one.

A gain is only a contribution if it can be shown statistically, so the data was resampled by call, three thousand times. The 95 % confidence interval on the gain is $[0.063, 0.104]$; no resample produced a non-positive gain, so $p < 1/3000$, which is the finest bound three thousand resamples can support. Resampling by call rather than by silence matters, because silences from one call are not independent and treating them as such would inflate the significance. The gain is not noise.

The fitted weights are worth reporting, since Section 3.6 claims interpretability as a reason for choosing this form. Averaged over the five folds they are $w_0 = -3.59$, $w_{\text{ac}} = 1.95$, $w_{\text{lex}} = 4.29$ and $w_{\text{inter}} = 1.22$, with a standard deviation across folds of at most 0.21. The lexical branch therefore carries about 2.2 times the weight of the acoustic one, which agrees with the two branches scored separately and is the opposite of what the framing of this project assumed.

The bottom of the table sets the condition under which fusion helps at all. The lexical signal only helps when it is in distribution. The medical-fitted model, applied to the generic French of the benchmark, degrades the decision to 0.789, below the acoustic baseline it was meant to improve. Fusion is not beneficial in itself. It is beneficial on condition that the lexical model matches the data, and that condition bounds where this method can be deployed.

The comparison against the other systems needs care. The table adds ultraVAD, an open-source detector that scores on audio alone, like Smart Turn. At 0.817 it edges past the acoustic baseline. That is worth recording, because it is the only comparator here that is neither commercial nor part of this project. At 0.897 the fusion sits above it and above LiveKit v1-mini (0.884), but below Deepgram Flux (0.915) and LiveKit v1 (0.937). It does not beat the strongest commercial detectors on AUC, and the margin against v1-mini is too small to claim a win there either.

On latency it trails as well: 715 ms against 557 ms for Soniox and 635 ms for LiveKit v1 at a 5 % budget. Deciding at a micro-pause rather than as the text arrives is a real cost, and a deliberate consequence of not blocking on transcription (Section 3.5). It is not the whole explanation, though: two of the three streaming services are slower still, AssemblyAI at 822 ms and GPT-Realtime 2 at 837 ms.

Figure 5.1 extends the two budgets of the table to the whole frontier. The fusion curve sits below the acoustic one across the entire range, so the gain is not an artefact of the operating points chosen for reporting. It also sits above the LiveKit curve throughout, so neither is the shortfall.

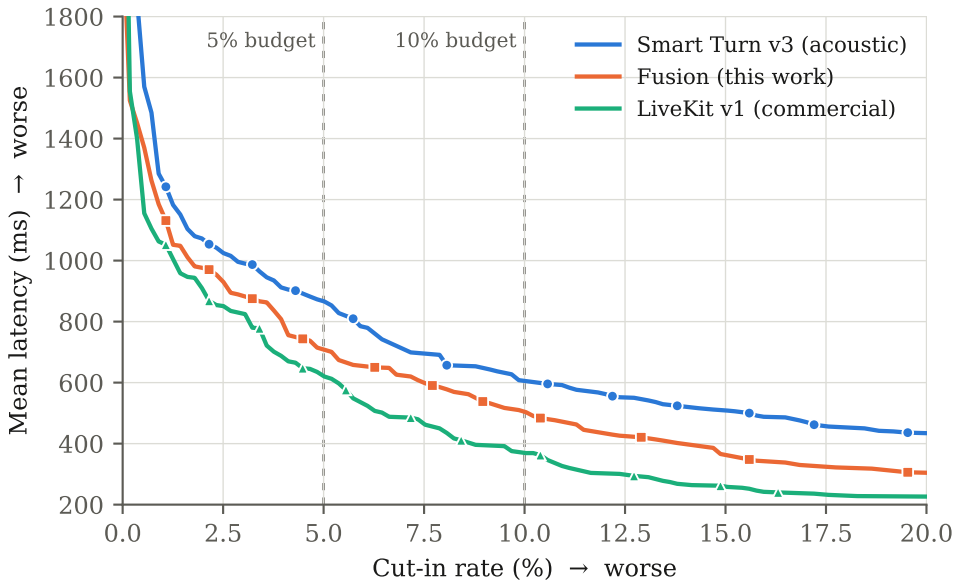


Figure 5.1: Swept latency-versus-cut-in frontier on the French benchmark. Lower and to the left is better. The dashed lines mark the 5 % and 10 % cut-in budgets quoted in Table 5.1.

5.3 Where the gain comes from

The project is motivated by three cases where silence is a poor cue: a name being spelled out, digits given in blocks, a pause on a date. It seemed worth checking whether the gain actually lands on those cases. Each silence was tagged from the causal caller text preceding it, and the AUC recomputed within each subgroup (Table 5.2).

Table 5.2: Per-case breakdown on the French benchmark. Subgroup sizes are given first because several groups are far too small to support inference: the spelling and date rows rest on six and four spans, so their AUCs are reported for completeness only and no weight is placed on them. Only the digit and plain-text rows are large enough to compare.

Case	n	eot	hold	AUC acoustic	AUC fusion
Spelling	6	1	5	0.800	1.000
Digits	269	107	162	0.825	0.875
Date	4	2	2	1.000	1.000
Plain text	764	290	474	0.807	0.901
Empty transcript	11	0	11	n/a	n/a
All	1,054	400	654	0.814	0.897

The first of the two findings runs against the framing of this project. On the digit case, which is large enough to trust at $n = 269$, fusion gains 0.050; on ordinary text it gains 0.094. The motivating case is the one where the lexical signal helps *least*.

That is explicable in hindsight: a partially dictated number is lexically ambiguous in a way an unfinished sentence is not, because nothing in the words themselves says whether more digits are coming. The hard cases stay hard, and the bulk of the headline gain comes from ordinary utterances.

The second is about the benchmark rather than the method itself. Spelling appears six times in 1,054 silences and dates four times. Generic French, as sampled here, essentially does not contain the phenomena that make medical intake difficult, so the public benchmark cannot test the case the project exists for. This is an argument for in-domain evaluation, not against it, and it is the reason Section 5.7 matters even though its result is negative.

5.4 Calibration

Table 5.3 gives the expected calibration error of Equation 3.2, before and after post-hoc isotonic recalibration, measured on a held-out split.

Table 5.3: Expected calibration error, raw and after post-hoc isotonic recalibration on a held-out split. Every external system that can be scored at the evaluation point is listed; nothing is selected. The raw column is the probability each system actually emits, and the calibrated column is what an integrator can reach by correcting it, which is available to every row equally.

Model	ECE (raw)	ECE (calibrated)
Smart Turn v3	0.216	0.079
ultraVAD	0.108	0.071
LiveKit Turn Detector v1	0.091	0.052
Deepgram Flux	0.087	0.083
LiveKit Turn Detector v1-mini	0.084	0.051
This work (fusion)	**0.064**	0.043

The three streaming services are excluded here. An expected calibration error pairs a predicted probability with the outcome of the same span, which is exactly the pairing Section 5.1 rejects for their AUC; reporting one while refusing the other would be inconsistent. One figure from those excluded runs is worth quoting, not as a result but as evidence for the exclusion itself: sampled at the

evaluation point, GPT-Realtime 2 falls from an ECE of 0.374 to 0.007 under recalibration. A collapse that large is what a near-constant score looks like, and a system that has not yet decided emits exactly that. No row is given for it in Table 5.3.

This chapter treats ECE > 0.05 as miscalibrated. The line is a working convention, not a standard. All five external detectors that can be scored at the evaluation point fall above it, as do ten of the eleven runs in the span set. `calibration_study.py` in the submitted corpus prints every run.

Before any correction, the fused probability is the best calibrated of the set, at 0.064 against 0.084 for the nearest external system. That is the comparison that bears on deployment, since none of these systems ships a recalibration step and an integrator therefore gets the raw column. The margin should not be pressed. v1-mini, Deepgram Flux and LiveKit v1 are all within 0.03, the table rests on a single held-out split of 214 spans, and no interval is attached to any of these values. The ordering among the four leading systems is not a result this dissertation can defend.

The calibrated column does not favour this work. Recalibration needs only a held-out split, so any integrator can apply it to any row. Applied uniformly it compresses the range: v1-mini reaches 0.051 and LiveKit v1 0.052 against 0.043 here. The ordering survives, the distance does not. Across all eleven runs the best-calibrated is this project's own dialogue-context variant at 0.039, which Section 5.6 reports as a failure on every other measure. The finding this section supports is therefore about the field and not about a ranking: miscalibration is universal, correcting it is cheap, and nobody does it. Nor does that spare this work, which sits uncorrected at 0.064, above the same line.

The mechanism is the one described in Section 2.3: these are neural classifiers trained to discriminate. Nothing in a discriminative objective rewards a probability that matches a frequency, and the leaderboard never asks for one.

The consequence is practical. A threshold placed on a probability that does not correspond to a real frequency cannot be connected to a cost, and every one of these systems exposes a threshold to its users as though it could be.

Because isotonic regression is monotonic, this correction moves the probability values without touching their ordering, which is why none of the AUCs in Table 5.1 change under it.

5.5 The cost-derived operating point

With a calibrated probability the stopping rule can select its operating point from the cost ratio λ of Equation 3.3. Table 5.4 shows how the point moves.

Table 5.4: Operating point as a function of the cost ratio $\lambda = C_{\text{cut}}/C_{\text{lat}}$.

λ	Cut-in rate	Latency
1 (one interruption = 1 s of latency)	16.8 %	330 ms
5	5.4 %	674 ms
10	1.8 %	981 ms
50 (one interruption = 50 s of latency)	0.2 %	1525 ms

The endpoint follows from the cost ratio, and the 500 ms of the existing system appears nowhere in the calculation. The behaviour is monotone and readable: the more costly an interruption is judged to be, the more conservative the endpoint becomes.

For medical telephone intake, $\lambda = 10$ is the setting this work would put forward: 1.8 % cut-in at 981 ms. Section 3.8 puts the cost of one interruption at about ten seconds, roughly the length of a repair turn when a caller cut off mid-identifier has to be asked again. Ten seconds plainly outweighs the 307 ms saved by moving to $\lambda = 5$. That rests on judgement: the cost of a repair turn was never measured, and measuring it on production dialogues is the proper way to fix λ. What the framework provides is the means to act on such a cost once known.

How the selected policies reach those points is worth recording, because it bears on the limits of a threshold rule. The score threshold is not monotone in λ: across the four rows it moves 0.37, 0.81, 0.32, 0.46. What does move monotonically is the timing. The action delay stays at 0.2 s for $\lambda = 1$ and 5, then rises to 0.9 s and 1.0 s, and the timeout goes from 1.0 s to 1.5 s and then 3.0 s. At high cost ratios the sweep buys conservatism by waiting longer, not by demanding a higher score, and it even lowers the threshold to do so. A rule that only moves a threshold on the current probability cannot reproduce that, which is what motivates the sequential formulation in Chapter 6.

5.6 Negative results

Three things were tried and did not work. They are reported because they mark the edges of the method. The first, a medical-fitted lexical read on generic French, is the 0.789 row of Table 5.1 and was discussed in Section 5.2; the other two follow.

Folding the elapsed silence into the score did not beat the harness sweep. This variant does not threshold the acoustic probability directly. It refits g on the acoustic probability and the silence duration d, over the terms p_{ac}, d, $p_{\text{ac}}d$ and d^2, so one threshold on g acts as a time-varying threshold on the original score. Applied to the acoustic baseline it moved the AUC from 0.814 to 0.811. The frontier is bounded by the quality of the score. The sweep already extracts the best temporal policy, so folding time into the probability adds nothing it was not already finding.

Augmenting the lexical input with dialogue context did not help either, dropping from 0.897 to 0.854. The completion signal lives in the caller's own words, and mean-pooling frozen embeddings over the agent's turn as well dilutes it.

On generic French, then, the method plateaus around 0.897. Going further needs either larger models, outside the scope of this project, or in-domain data, which the next section tests.

5.7 In-domain validation

Everything above runs on a clean, generic, public dataset. The in-domain validation uses French medical calls captured by instrumenting production. Of the 709 calls captured (Section 4.6), 636 contribute at least one scored micro-pause. Those calls yield 5,342 micro-pauses: 1,015 ends of turn and 4,327 holds. The labels come from weak supervision. The recordings are stereo, caller on the left channel, sampled at 8 kHz, the standard telephone band.

Contrary to my hypothesis, fusion does not win in domain. Per pause the AUCs are 0.49 for the acoustic branch alone, 0.60 for the lexical branch alone and 0.59 for the fusion. The acoustic model is at chance and contributes nothing. Worse for the argument of this dissertation, fusing is actively harmful here: 0.59 sits below the 0.60 of the lexical branch on its own, so combining a useful signal with a useless one costs more than it returns. That is the direct counter-example to the benchmark result of Section 5.2, and it holds under the same condition stated there, that fusion helps only when both inputs carry signal.

Two causes explain this, once bugs are ruled out. The caller channel was confirmed to be the left one by measuring speech energy in the audio preceding each pause, and both the timestamp alignment and the scoring point were checked against the benchmark path.

The first cause is bandwidth, and it can be measured rather than assumed. The recordings are sampled at 8 kHz, so by Nyquist they carry nothing above 4 kHz; across sixty of them, 0.32 % of the energy sits above the 3.4 kHz telephone cut-off

and 0.007 % above 4 kHz. Smart Turn is trained on 16 kHz wideband, so half the spectrum it learned on is simply absent.

The effect on its output is visible in the distribution. On the benchmark the score is spread across the range, with an interquartile range of 0.94 and 41 % of spans scored below 0.1. In domain it collapses against the ceiling: median 0.98, interquartile range 0.09, and 75 % of pauses scored above 0.9. The separation between the two classes disappears with it. On the benchmark the mean score is 0.738 for ends of turn against 0.287 for holds, a gap of +0.45; in domain it is 0.820 against 0.844, a gap of −0.02. The model scores holds marginally higher than ends of turn, which is what an AUC of 0.49 means. A generic acoustic detector does not transfer to 8 kHz telephony.

The second is the labels. The comparison to draw is between the same lexical branch on the two datasets. On the public benchmark, where boundaries are human-verified, a frozen CamemBERT embedding under a logistic head reaches 0.845. The identical construction on the in-domain set, with labels inferred from resume behaviour, plateaus near 0.60. Nearly a quarter in absolute AUC separates them, and the only thing that changed is where the labels came from. Whether a patient resumes speaking is weakly predicted by the words uttered so far, because a pause followed by a resume and a final pause often have near-identical preceding text. The ceiling is in the supervision, not in the model.

The per-case breakdown is reported on the public benchmark rather than here. The reason is measurement validity. The in-domain set is roughly five times larger, but with the acoustic branch at chance and the lexical signal capped at 0.60, splitting an uninformative measurement by sub-case would yield nothing interpretable. A breakdown in domain requires the manually annotated subset identified in Chapter 6, which the project timescale did not allow.

So the in-domain validation does not confirm that the method transfers. It shows where it breaks. This does not contradict the headline result, which stands on clean labels, but it does show that reaching production takes adaptation. Three directions follow, taken up in Chapter 6: annotate a sample by hand to obtain a clean in-domain test set; adapt an acoustic model to the 8 kHz telephone band; rebuild the causal text from per-turn final transcripts instead of concatenated partial hypotheses.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

This project asked a simple question: does adding a semantic signal to an acoustic one, and then deriving a cost-sensitive decision from the result, improve end-of-turn detection for a French medical voice agent? On a clean public benchmark the answer is yes. Fusion raises the AUC from 0.814, for the acoustic model alone, to 0.897, a statistically significant gain of 0.083, and neither base model is fine-tuned in that pipeline. A medically fine-tuned lexical model exists and serves the in-domain path and the live demonstration, but it is not what produces this number.

What the project learned is worth as much as what it built. Fusion only helps when the lexical model is in distribution: applied to generic French, a medically tuned model degrades the decision below the baseline it was meant to improve. Calibration goes unreported, though a calibrated probability is the precondition for deciding from a stated cost rather than an arbitrary timer: every external detector evaluated here is miscalibrated on this data and none publishes any measurement of it. And the off-the-shelf pipeline does not transfer to production, for reasons Section 5.7 sets out and O4 returns to.

There is one more finding I did not expect, and it cuts against the framing of the project. Section 5.3 shows the fusion gaining least on precisely the cases the work exists for, and shows that generic French barely contains those cases at all. The motivating problem is real, but the public benchmark cannot measure it, and the lexical signal is weakest exactly where I argued it would be strongest.

This dissertation contributes a calibration-first, cost-aware method for composing existing models, tested in a real setting, and it measures where that method stops working.

6.2 Evaluation against the objectives

O1, a modular real-time pipeline whose decision never blocks on the transcript, is met: the package is in place, covered by 45 automated tests, and demonstrated end to end from the command line and in the browser.

O2, fusing off-the-shelf acoustic and lexical models and testing whether this beats the acoustic model alone, is met (Section 5.2).

O3, calibrating the probability and deciding from a cost, is met as analysis and not as a property of the running system. Calibration was measured across the whole leaderboard, and the operating point is derived from the cost ratio rather than from a fixed delay, but that derivation lives in the evaluation: the shipped controller still maps confidence onto a hand-set silence ladder.

O4, instrumenting production and validating in domain, was carried out, and its result is negative. The capture works, with 709 calls recorded and 636 scored. The eot-bench converter was written but never exercised end to end, because the harness route was blocked by a dependency conflict (Section 4.5), so the in-domain figures come from a standalone scorer instead. The validation showed that the off-the-shelf pipeline does not transfer: the acoustic branch is at chance on 8 kHz audio and the lexical branch plateaus near 0.60 on weak labels. The objective was delivered as engineering and as measurement; the hypothesis it was built to test was not confirmed.

Several limitations bound what has been shown. On generic French the method plateaus around 0.897, and passing that needs larger models, which is outside the contribution. The system trails the fastest commercial detectors on latency, because it waits for a micro-pause rather than deciding as the text arrives, though two of the three streaming services are slower still. The in-domain labels are weak, and the French benchmark subset is small at roughly a thousand silences.

There are things I would do differently. I would instrument production in the first week rather than the seventh, so that a properly annotated in-domain set existed in time instead of a weakly supervised one. I would check the audio bandwidth of the target deployment before choosing an acoustic model. The 16 kHz-to-8 kHz gap surfaced at validation, which is far too late for it to surface. And I would run the per-case breakdown early: had I known in week three that the public benchmark contains six spelling cases, I would have known from the start that it could not test the thing the project was for.

The three pillars of this work were set out before implementation began, in the critical review described in Chapter 1, and two of them survived intact. The third bent: I evaluate at the micro-pause rather than continuously as speech arrives, which is exactly the latency gap of Section 5.2 and the first item of future

work below. Being overtaken mid-project was the other thing that shaped the contribution. Narrowing the claim to calibration and cost, instead of competing on fusion itself, was, I still think, the right response.

6.3 Future work

The first direction is latency, which is what stands between this system and deployment. It decides at the micro-pause, which is why it trails services that commit as the text arrives. Two levers exist: a streaming lexical model that scores word by word and can decide before the pause, and the stability of the recogniser's interim hypotheses, how quickly the transcript stops changing, as an early cue. That second signal is only observable in the partial-hypothesis trajectories captured in production, and cannot be tested on the public benchmark, which supplies final words only.

The second answers the in-domain result directly: adapt the acoustic model to the 8 kHz telephone band, by retraining or by starting from a natively narrowband model, to remove the saturation, and hand-annotate a few hundred pauses to obtain clean end-of-turn labels and a trustworthy in-domain measurement, which weak supervision cannot provide.

The third is a sequential optimal-stopping formulation. The current rule is myopic: it decides on the evidence available at one instant, applying a threshold to the calibrated probability without weighing what waiting a little longer might reveal. At high cost ratios the swept oracle turns conservative through timing rather than through the threshold, and a rule that only looks at the present instant cannot match that. A multi-step cost model, in the tradition of Wald (1945), would track the oracle across every cost regime.

Appendix A

Ethics and Data Handling

A.1 Context and partnership

This project was carried out in partnership with Tennor, which operates a French medical voice agent that patients call by telephone. The work builds a real-time end-of-utterance detector for that agent. Tennor is the data controller and operates a GDPR-compliant, ISO/IEC 27001-certified environment. I processed the data inside Tennor's infrastructure as part of the partnership. No participants were recruited and there was no interaction with patients: the work reuses signals already generated by the live system.

Correction to the statement submitted with the corpus. The ethics statement submitted alongside the code corpus on 21 August 2026 describes the capture as limited to *internal test organisations*. That is narrower than what actually happened: capture was feature-flag gated at organisation level on *production* organisations. This appendix supersedes that statement and gives the accurate position. The correction was raised with my supervisor before writing. The corpus itself can no longer be amended.

A.2 Data used

Public benchmark, no personal data. Every headline result reported in Chapter 5 runs on the LiveKit eot-bench dataset (CC-BY-4.0), a public French turn-taking benchmark containing no Tennor data and no personal data. The benchmark deliberately carries the weight of the evaluation, so that the method can be demonstrated without personal data at all (Section 3.2).

Production in-domain set. A secondary, in-domain set was derived from calls to the Tennor agent: 709 calls placed on a single production organisation, of which roughly four in five are genuine inbound calls and the remainder calls made by

internal testers to the same line. It is used only as an in-domain validation set, and it is not included in the corpus.

A.3 Legal basis and agreement with the controller

Tennor is the data controller. Every use of Tennor data in this project was agreed with the company before it took place, and the company reviewed the scope of the capture, the fields retained, this appendix, and the dissertation itself. Tennor is aware of what the work covers and of what it implies for the production agent. Nothing was collected, processed, or released outside that agreement, and the partnership was conducted transparently on both sides.

One consequence follows directly. The in-domain production set described above, referred to as the *gold* set in the submitted corpus, cannot be shared or made public. The reason is data sovereignty: these are French medical call recordings and transcripts held in Tennor's own infrastructure, and they stay there. The set is therefore described and measured in this dissertation but not distributed with it, which is why the public benchmark carries every headline result (Section 3.2).

A.4 Pseudonymisation, not anonymisation

Names, first names and ages are masked in the retained transcripts, and the export is reduced to the fields the task uses. The data nonetheless remains personal data under the GDPR: the caller-side audio is retained and every record is linked to a production call identifier, so re-identification by the controller remains possible. It is treated as personal data throughout, and describing it as anonymised would be incorrect. The fields the task uses are the caller text, the recogniser's interim trajectory, the pause and decision timestamps, an audio reference, and the resume and interruption flags. The export is reduced to these.

A.5 Reduced human exposure

Labels are produced automatically by weak supervision, from whether the caller resumed speaking after a pause (Section 4.6). Building the dataset therefore required nobody to sit and listen to call content, which limits human exposure to what patients said. The cost of that choice is label noise, and it bounds the result reported in Section 5.7. It is an ethical decision with a measurable methodological price, which is worth stating in both directions.

A.6 Storage, access and the corpus

The data and any derived audio or transcripts stay within Tennor's infrastructure. Access was restricted to me. Nothing was copied into the repository, the corpus, or any public location: `.gitignore` excludes call audio, transcripts and model weights, and the corpus was packaged and checked so that none of those files were included. The corpus contains the source code, the dataset-generation code, the evaluation on the public dataset, the design documents and this statement.

A.7 Risks and mitigations

Risk	Mitigation
Exposure of patient data	Nothing leaves Tennor's infrastructure; no audio, transcript or derived record in the corpus; access limited to the author
Re-identification	Names, first names and ages masked in the retained transcripts; treated as personal data throughout rather than described as anonymised; nothing exported beyond Tennor's infrastructure
Accidental inclusion in the corpus	`.gitignore` rules plus packaging that excludes audio, transcripts and weights, verified before submission
Over-collection	Only caller-side, task-relevant fields captured per micro-pause
Human exposure to call content	Automatic weak-supervision labelling instead of manual listening
Clinical impact of a wrong cut	Cost-sensitive decision with the interruption cost stated explicitly, and cut-in rates reported openly; the system augments rather than replaces existing safeguards

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